

claude-windows / c1fcb4aa-537d-41fd-858e-53f93349bf5a

Metadata

```
- Source: `claude-windows`  
- Kind: `claude`  
- Source file: `C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\c1fcb4aa-537d-41fd-858e-53f93349bf5a\subagents\workflows\wf_41c8dc8a-8a4\agent-aa0e778fece4ff079.jsonl`  
- SHA-256: `6933a942046f63c4c30e18f2bd567f6b6160f7bdcacfe09f5e5792275e55593b`  
- Source modified: `2026-07-02T02:53:14+00:00`  
- Imported at: `2026-07-05T16:48:24+00:00`  
- project: `wf_41c8dc8a-8a4`  
- session_id: `c1fcb4aa-537d-41fd-858e-53f93349bf5a`
```

Transcript

1. user (2026-07-02T02:47:11.979Z)

CONTEXT (read carefully):

- You are one auditor in a multi-agent tech-debt/critical-fix audit of the khelsutra-guru multi-repo workspace at C:/Users/avidu/Projects/khelsutra-guru. Today is 2026-07-02. All git repos were just pulled to current origin master/main.
- Repos: badminton-highlight-indexer (the ENGINE, Python, ~1500 tests, org Khelsutra), khelsutra (SPA web/ + marketing site/, TS/React, org avidullu), sports-data-collector, rally-annotator, sports-obsreport (small Python repos). Plus two NON-git dirs: "Annotation Setup", khelsutra-screencast.
- HARD RULES: READ-ONLY. Do NOT modify/create/delete any file, do NOT create branches/worktrees, do NOT pip install, do NOT run the full test suite (CI is trusted on master). You MAY run read-only git and gh commands (gh is authenticated), ruff/mypy IF already installed, and fast read-only scripts.
- EVERY claim must cite file:line (or a gh URL). A claim without file:line evidence is a hypothesis - mark it as such or drop it. The codebase drifts between sessions: re-verify anything you believe from docs against actual code.
- OWNER-GATED areas (flag as owner_gated=true, still report): alpha/serving go-live + Cloudflare dashboard work, Studio P10/P11 (corpus promotion + train/reeval), product/strategy/licensing decisions, real GPU/cloud spend, counsel ToS/DPA, repo rename off "badminton".
- Known-open items from project memory (VERIFY current state, don't trust blindly): engine PR #390 (D1 split main.py, open since 06-26) · audit doc docs/CODE_CLEANUP_AUDIT_2026-06-24.md remaining items D1, D13 (ByteTrack->trackers), Phase-3 (B3/B4/B6/B7/B8/D5/D6/D8-D10/03/04) · docs/DECISION_SNAPSHOT_REGRESSION.md next=P1 · docs/GOLDEN_REGRESSION_FIXTURES.md backlog M4/P0-rename/01/02 · engine issues #340 (Gemini re-bill), #332 (NSFW preflight), #349 (CPU-bound decode), #384 (CI single-source) · khelsutra issues #97-#114 UX/alpha backlog · DO droplet RETIRED 2026-06-25 (docs may still reference it; api.khelsutra.guru CNAME/CF-Access cleanup was pending).
- Categorize each finding: critical-fix (latent bug/correctness/data-loss/cost-leak), security, tech-debt (structure/duplication/dead code), test-gap, ci-infra, doc-rot, hygiene.
- Severity: P0 = latent bug or cost/data/security risk reachable in normal use; P1 = high-value debt blocking velocity or a stale-open thread; P2 = worthwhile cleanup; P3 = nice-to-have (report max 3 of these).
- Be selective: max ~12 findings, each genuinely worth acting on. Note tracked_in when an issue/PR/doc already tracks it (tracked is NOT done - still report). Your final message is data for an orchestrator, not prose for a human.

YOUR DIMENSION: CI + test health of the ENGINE. Repo: C:/Users/avidu/Projects/khelsutra-guru/badminton-highlight-indexer.

Check: gh run list --limit 15 (any red/flaky on master?); .github/workflows/*.yaml - duplication that issue #384 tracks (shared offline-guard lanes + coverage floor via workflow_call - verify current state), per-job timeouts, nightly.yml health (gh run list --workflow=nightly.yml --limit 10 - is the Tier-1 nightly green or silently red?); pytest config + coverage floor (pyproject/setup.cfg --cov-fail-under vs actual); count of skipped tests and why (grep for skipif/skip markers - are the 12 obsreport skips still the only ones?); slowest-test hygiene;

ruff/mypy config drift. Report red/flaky CI as P0/P1.

2. user (2026-07-02T02:47:20.146Z)

e410136 api: /api/videos surfaces fps + resolution (from the stored ffprobe details) (#413)
81a78bf api: /api/videos returns video_id + a friendly match_name (fix disappearing name) (#412)
8b16cf4 docs: reflect today's storage endpoints + the corpus/user-video boundary (#411)

master...origin/master

3. user (2026-07-02T02:47:21.189Z)

.github\workflows\nightly.yml
.github\workflows\ci.yml

4. user (2026-07-02T02:47:29.690Z)

completed	success	api: /api/videos surfaces fps + resolution (from the stored ffprobe d?			
CI	master	push	28532862647	2m58s	2026-07-01T16:36:43Z
completed	success	api: /api/videos returns video_id + a friendly match_name (fix disapp?			
CI	master	push	28531005211	2m53s	2026-07-01T16:04:25Z
completed	success	docs: reflect today's storage endpoints + the corpus/user-video bound?			
CI	master	push	28530352910	2m40s	2026-07-01T15:53:37Z
completed	success	deploy/gcp: centralize project + bucket names in one sourced buckets.?			
CI	master	push	28528006637	3m0s	2026-07-01T15:15:59Z
completed	success	P4: POST /api/assets ? store a video into the user's chosen medium (#?			
CI	master	push	28524662552	2m45s	2026-07-01T14:23:42Z
completed	success	P3: /api/capabilities reports usable storage backends (#408)		CI	
master	push	28523306298	2m50s	2026-07-01T14:02:36Z	
completed	success	P2: engine free-space guard (HTTP 507 + WSL frame-stage BLOCK) (#407)			
CI	master	push	28523034384	2m37s	2026-07-01T13:58:28Z
completed	success	feat(api): frame-preserving annotation proxy for /api/source (D4) (#406)			
CI	master	push	28519443869	2m48s	2026-07-01T13:01:41Z
completed	success	feat(api): P8a ? GET /api/source + POST /api/annotations (annotate-in?			
CI	master	push	28503170582	2m49s	2026-07-01T08:09:11Z
completed	success	Nightly quality regression (Tier-1, offline)		Nightly quality	
regression (Tier-1, offline)	master	schedule	28499884740	2m48s	
2026-07-01T07:05:07Z					
completed	success	Nightly quality regression (Tier-1, offline)		Nightly quality	
regression (Tier-1, offline)	master	schedule	28425848895	2m42s	
2026-06-30T06:49:07Z					
completed	success	docs: record the launch NO-GO decision (owner, 2026-06-30) (#404)			
CI	master	push	28419282606	2m56s	2026-06-30T03:59:24Z
completed	success	docs: fix stale rally-cluster claims (no archival ? cluster still gat?			
CI	master	push	28417968091	2m37s	2026-06-30T03:20:56Z
completed	success	chore(#399): launch-PR process hardening ? link-check CI, PR template?			
CI	master	push	28417781155	2m31s	2026-06-30T03:15:23Z
completed	success	fix(#398): served Gate-A litmus passes a typed IndexingConfig (not a ?			
CI	master	push	28417374248	2m52s	2026-06-30T03:03:56Z

5. user (2026-07-02T02:47:31.069Z)

completed	success	Nightly quality regression (Tier-1, offline)		Nightly quality	
regression (Tier-1, offline)	master	schedule	28499884740	2m48s	
2026-07-01T07:05:07Z					
completed	success	Nightly quality regression (Tier-1, offline)		Nightly quality	
regression (Tier-1, offline)	master	schedule	28425848895	2m42s	
2026-06-30T06:49:07Z					
completed	success	Nightly quality regression (Tier-1, offline)		Nightly quality	
regression (Tier-1, offline)	master	schedule	28355510707	2m37s	
2026-06-29T07:22:43Z					
completed	success	Nightly quality regression (Tier-1, offline)		Nightly quality	
regression (Tier-1, offline)	master	schedule	28314418544	2m45s	
2026-06-28T06:56:14Z					
completed	success	Nightly quality regression (Tier-1, offline)		Nightly quality	

regression (Tier-1, offline)	master	schedule	28281089675	2m45s
2026-06-27T06:23:13Z				
completed	success	Nightly quality regression (Tier-1, offline)	Nightly quality	
regression (Tier-1, offline)	master	schedule	28222002166	2m30s
2026-06-26T06:44:51Z				

6. user (2026-07-02T02:47:31.094Z)

```

1      name: CI
2
3      on:
4        push:
5          branches: [master]
6        pull_request:
7
8      permissions:
9        contents: read
10
11     concurrency:
12       group: ci-${{ github.ref }}
13       cancel-in-progress: true
14
15     jobs:
16       # Lint gate (ruff.toml at repo root): pyflakes/pycodestyle + bugbear.
17       # Also the repo-wide syntax guard ? ruff reports invalid-syntax on every
18       # tracked .py, including scripts/ and training/, which the backend-only
19       # py_compile smoke sweep does not reach.
20       lint:
21         name: Ruff lint
22         runs-on: ubuntu-latest
23         timeout-minutes: 10
24         steps:
25           - uses: actions/checkout@v6
26           - uses: actions/setup-python@v6
27             with:
28               python-version: "3.13"
29           - name: Install ruff
30             run: python -m pip install ruff
31           - name: Ruff check
32             run: ruff check . --output-format github
33
34       # PII / attribution leak guard (tools/leak_scan.py): fails on a committed 32-hex MD5
35       # (the video_id shape) or a personal absolute user path in the shipped-code surface
36       # (backend/training/scripts/tools/eval_baselines; tests/ excluded ? legit fixtures).
37       Real
38       # names are scanned only via a gitignored private list locally; CI runs the structural
39       guard.
40       leak-scan:
41         name: Leak scan (PII / attribution guard)
42         runs-on: ubuntu-latest
43         timeout-minutes: 10
44         steps:
45           - uses: actions/checkout@v6
46           - uses: actions/setup-python@v6
47             with:
48               python-version: "3.13"
49           - name: Leak scan
50             run: python -m tools.leak_scan
51
52       # Internal-link guard (tools/check_md_links.py): fails on a broken internal markdown
53       link
54       # (missing file or dead #anchor) in a LIVE doc. docs/archives/** is excluded ? frozen
55       # snapshots whose relative links are historical-by-design (see docs/DOC_STATUS.md §2).
56       link-check:
57         name: Markdown link check (live docs)

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55     runs-on: ubuntu-latest
56     timeout-minutes: 10
57     steps:
58     - uses: actions/checkout@v6
59     - uses: actions/setup-python@v6
60       with:
61         python-version: "3.13"
62     - name: Check internal markdown links
63       run: python -m tools.check_md_links
64
65     # Type gate (mypy.ini at repo root): lenient green baseline with an explicit
66     # per-module quarantine list ? ratchet by removing entries as modules clean
67     # up. Typed core deps are installed so mypy sees their real signatures
68     # (without them ignore_missing_imports would Any-stub half the checks).
69     typecheck:
70       name: Mypy (lenient baseline)
71       runs-on: ubuntu-latest
72       timeout-minutes: 10
73       steps:
74       - uses: actions/checkout@v6
75       - uses: actions/setup-python@v6
76         with:
77           python-version: "3.13"
78       - name: Install mypy + typed core deps
79       run: python -m pip install mypy pydantic fastapi uvicorn python-dotenv google-
genai numpy
80       - name: Mypy
81       run: mypy backend training
82
83     # Fast guard against the 7fbb0 class of breakage: a SyntaxError or
84     # module-level import error shipping while the suite stays green.
85     # Deliberately does NOT install cv2/torch/numpy ? the smoke tests must hold
86     # on machines without the GPU stack (test_import_smoke stubs what's absent).
87     import-smoke:
88       name: Syntax sweep + import smoke (no GPU stack)
89       runs-on: ubuntu-latest
90       timeout-minutes: 10
91       steps:
92       - uses: actions/checkout@v6
93       - uses: actions/setup-python@v6
94         with:
95           python-version: "3.13"
96       - name: Install minimal deps
97       run: python -m pip install pytest fastapi uvicorn pydantic python-dotenv google-
genai
98       - name: Run import smoke tests
99       run: python -m pytest tests/test_import_smoke.py -v
100
101     full-suite:
102       name: Full test suite (offline, mocked)
103       runs-on: ubuntu-latest
104       timeout-minutes: 30
105       steps:
106       - uses: actions/checkout@v6
107       - uses: actions/setup-python@v6
108         with:
109           python-version: "3.13"
110           cache: pip
111       - name: System libs for opencv-python
112       run: |
113         # Drop the runner's pre-installed Microsoft apt repo (azure-cli): it
periodically
114         # 403s on InRelease and aborts `apt-get update` under the `-e` shell, failing
the
115         # build on an unrelated repo we don't use. libgl1/libglib2.0 are in Ubuntu's

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archive.
116         sudo rm -f /etc/apt/sources.list.d/*microsoft* /etc/apt/sources.list.d/*azure*
117         sudo apt-get update
118         sudo apt-get install -y libgl1 libgl1-mesa-dev || sudo apt-get install -y
libgl1 libgl1-mesa-dev
119     - name: Install CPU-only torch (no CUDA wheels on a CPU runner)
120       # torch/torchvision are NOT in pyproject deps ? their CPU/CUDA wheels
121       # need an out-of-band index that PEP 621 can't express. Keep installing
122       # them separately here, before the editable install pulls the rest.
123       run: python -m pip install torch torchvision --index-url
https://download.pytorch.org/whl/cpu
124     - name: Install trackers (git-only dep ? PyPI wheel is broken)
125       # D13/P12: ByteTrack moved out of `supervision` (removed in 0.30.0) to the
126       # standalone `trackers` package. Its PyPI wheel ships metadata only (no
127       # source ? roboflow/trackers#474), so it MUST come from git. PEP 621 can't
128       # express a git URL, so it's not in pyproject.toml (same pattern as torch).
129       # Installed here so the two D13/P12 tests in tests/test_person_detector.py
130       # actually run in CI instead of skipping via pytest.importorskip.
131       # NOTE: roboflow/trackers uses bare version tags (2.5.0), NOT v-prefixed.
132       run: python -m pip install "trackers @
git+https://github.com/roboflow/trackers.git@2.5.0"
133     - name: Install package (editable) + dev tooling
134       # Editable PEP 621 install: runtime deps + the `dev` group
135       # (pytest, pytest-cov, httpx for the fastapi TestClient, ruff, mypy).
136       run: python -m pip install -e .[dev]
137       # obsreport (private, soft dep) is absent here: reporting tests skip via
138       # pytest.importorskip ? run_all_tests.py prints which. The coverage floor
139       # is therefore calibrated against THIS lane's number (local runs with
140       # obsreport installed measure a little higher).
141       # Floor calibrated 2026-06-11 against this lane's measured 72% (local
142       # with obsreport: 73%). Ratchet it UP as coverage grows; never down
143       # without an owner decision.
144     - name: Run full suite (with coverage floor)
145       run: python run_all_tests.py --cov=backend --cov=training --cov-report=term
--cov-fail-under=70
146
147     # Cross-OS determinism guard for the committed golden regression fixtures
148     # (docs/GOLDEN_REGRESSION_FIXTURES.md). The CONTROL replay path is pure-stdlib
149     # (only numpy, no cv2/torch), so this is a FAST, focused job ? it does NOT run
150     # the full suite x3 OSes. It replays the Windows/Linux/macOS runner against the
151     # committed (parse-compared) snapshots, catching any cross-OS float drift beyond
152     # the 1e-6 tolerance before it can flake a contributor's PR on another machine.
153     golden-crossos:
154       name: Golden fixtures cross-OS (${{ matrix.os }})
155       strategy:
156         fail-fast: false
157         matrix:
158           os: [ubuntu-latest, windows-latest, macos-latest]
159       runs-on: ${{ matrix.os }}
160       timeout-minutes: 15
161       steps:
162         - uses: actions/checkout@v6
163         - uses: actions/setup-python@v6
164           with:
165             python-version: "3.13"
166         - name: Install minimal deps (no GPU stack ? the CONTROL replay needs only numpy)
167           run: python -m pip install pytest fastapi uvicorn pydantic python-dotenv google-
genai numpy
168         - name: Golden-fixtures characterization (cross-OS determinism guard)
169           # Also runs the freeze-real / time-origin-reset / M2 refusal-gate tests
(synthetic /
170           # tmp_path only ? no GPU/network) so the reset metric-invariance (fix #2) and
the
171           # de-attribution paths get cross-OS coverage and surface any win/mac float
drift.

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172         run: |
173             python -m backend.eval.golden_fixtures --check
174             python -m pytest tests/test_golden_fixtures.py
tests/test_golden_real_fixtures.py -q
175
176     # Commercial-clean licensing guardrail (MIT/Apache/BSD only ? PACKAGING_PLAN.md P0d).
Scans the
177     # RUNTIME dependency closure (`pip install .` ? no torch/dev) and fails on any
copyleft
178     # (GPL/AGPL/LGPL/SSPL). torch (BSD-3) is installed out-of-band and is not in this
closure. This
179     # automates a guardrail that was human-only; the report is printed for review either
way.
180     # D13/P12 carve-out: `trackers` (Apache-2.0; ByteTrack's new home) is also git-only
and thus
181     # absent from this pyproject closure ? UNguarded here, same shape as torch. It's
Apache-2.0 and
182     # depends only on the still-declared `supervision` (MIT), so the closure is clean
today; if
183     # `trackers` ever pulls a new transitive dep, this gate won't catch it ? audit on
bump.
184     license-scan:
185         name: Dependency license scan (no copyleft in runtime deps)
186         runs-on: ubuntu-latest
187         timeout-minutes: 15
188         steps:
189             - uses: actions/checkout@v6
190             - uses: actions/setup-python@v6
191               with:
192                 python-version: "3.13"
193             - name: Install runtime deps + pip-licenses
194               run: |
195                 python -m pip install .
196                 python -m pip install pip-licenses
197             - name: Report dependency licenses
198               run: python -m piplicenses --format=markdown --with-urls --order=license
199             - name: Fail on any copyleft (GPL/AGPL/LGPL/SSPL) in the runtime closure
200               run: |
201                 if python -m piplicenses --format=plain | grep -iE "GPL|SSPL"; then
202                     echo "::error::A runtime dependency carries a copyleft (GPL/AGPL/LGPL/SSPL)
license ? the guardrail is MIT/Apache/BSD only. Review the report above."
203                     exit 1
204                 fi
205                 echo "OK: no GPL/AGPL/LGPL/SSPL in the runtime dependency closure."
206
207     # Built-wheel smoke (PACKAGING_PLAN.md P0d): build the wheel, install it NON-editable
from a
208     # non-repo cwd, and launch the console script. Catches packaging regressions the
source suite
209     # can't ? namespace-package resolution (backend has no __init__.py), missing data
files, hidden
210     # imports. (Validated locally: a clean-venv wheel install + CPU torch runs `indexer
--help`.)
211     build-artifact:
212         name: Built wheel smoke (install + launch)
213         runs-on: ubuntu-latest
214         timeout-minutes: 30
215         steps:
216             - uses: actions/checkout@v6
217             - uses: actions/setup-python@v6
218               with:
219                 python-version: "3.13"
220             - name: System libs for opencv-python
221               run: |
222                 # Drop the runner's pre-installed Microsoft apt repo (azure-cli): it

```

```

periodically
223         # 403s on InRelease and aborts `apt-get update` under the `-e` shell, failing
the
224         # build on an unrelated repo we don't use. libgl1/libglib2.0 are in Ubuntu's
archive.
225         sudo rm -f /etc/apt/sources.list.d/*microsoft* /etc/apt/sources.list.d/*azure*
226         sudo apt-get update
227         sudo apt-get install -y libgl1 libglib2.0-0t64 || sudo apt-get install -y
libgl1 libglib2.0-0
228     - name: Build the wheel
229       run: |
230         python -m pip install build
231         python -m build --wheel
232     - name: Install the built wheel + CPU torch (out-of-band index)
233       run: |
234         python -m pip install torch torchvision --index-url
https://download.pytorch.org/whl/cpu
235         python -m pip install dist/*.whl
236     - name: Launch the installed console script from a non-repo dir
237       run: |
238         cd /tmp
239         indexer --help
240         python -c "import backend, training, backend.utils.app_home; print('wheel
import OK')"
241

```

7. user (2026-07-02T02:47:31.095Z)

```

1     name: Nightly quality regression (Tier-1, offline)
2
3     # Tier-1 nightly quality guard. Per-PR `ci.yml` only fires on push/pull_request;
4     # this runs the SAME offline guards on a schedule so we also catch what per-PR CI
5     # can't: dependency / transitive-dep drift, flakiness over time, and rot during
6     # quiet periods when nothing is pushed.
7     #
8     # Tier-1 == OFFLINE ONLY: no GPU, no real video, no `run_regression.py` on the
9     # real corpus (that's Tier-2 ? owner-gated, needs a self-hosted GPU runner +
10    # the gitignored golden corpus).
11    #
12    # Lane parity with ci.yml: every OFFLINE guard job from ci.yml is mirrored here
13    # (lint · leak-scan · typecheck · import-smoke · full-suite · golden-fixtures ·
14    # license-scan · build-artifact), with the same Python pin + per-lane pip
15    # installs so modules collect/run instead of skipping for a missing dep. Two
16    # intentional differences, NOT drops:
17    #   - golden-fixtures runs single-OS here (ci.yml::golden-crossos already covers
18    #     the Win/mac/Linux matrix on every PR; the nightly is about drift-over-time,
19    #     not cross-OS float drift).
20    #   - a notify-on-failure job (below) surfaces a SCHEDULED failure as a tracking
21    #     issue ? ci.yml doesn't need it (a PR/push run already has a red check).
22    # Lane set + coverage floor are still copied literals, not single-sourced ? see
23    # the `workflow_call` single-sourcing follow-up in the PR description.
24    #
25    # NOTE: scheduled workflows only fire from the repo's DEFAULT branch, so this
26    # won't actually run on a cadence until it lands on `master`. Use the
27    # `workflow_dispatch` trigger to run it on demand for validation after merge.
28
29    on:
30      schedule:
31        # 03:00 UTC daily.
32        - cron: "0 3 * * *"
33      workflow_dispatch:
34
35    permissions:
36      contents: read
37

```

```

38 concurrency:
39   group: nightly- $\{\{ \text{github.ref} \}\}$ 
40   cancel-in-progress: true
41
42 jobs:
43   # Lint gate (ruff.toml at repo root) ? mirrors ci.yml::lint.
44   lint:
45     name: Nightly · Ruff lint
46     runs-on: ubuntu-latest
47     timeout-minutes: 10
48     steps:
49       - uses: actions/checkout@v6
50       - uses: actions/setup-python@v6
51         with:
52           python-version: "3.13"
53       - name: Install ruff
54         run: python -m pip install ruff
55       - name: Ruff check
56         run: ruff check . --output-format github
57
58   # PII / attribution leak guard (tools/leak_scan.py) ? mirrors ci.yml::leak-scan.
59   # A structural-rot guard: fails on a committed 32-hex MD5 (the video_id shape)
60   # or a personal absolute user path in the shipped-code surface.
61   leak-scan:
62     name: Nightly · Leak scan (PII / attribution guard)
63     runs-on: ubuntu-latest
64     timeout-minutes: 10
65     steps:
66       - uses: actions/checkout@v6
67       - uses: actions/setup-python@v6
68         with:
69           python-version: "3.13"
70       - name: Leak scan
71         run: python -m tools.leak_scan
72
73   # Type gate (mypy.ini at repo root) ? mirrors ci.yml::typecheck. Typed core
74   # deps are installed so mypy sees real signatures (without them
75   # ignore_missing_imports would Any-stub half the checks).
76   typecheck:
77     name: Nightly · Mypy (lenient baseline)
78     runs-on: ubuntu-latest
79     timeout-minutes: 10
80     steps:
81       - uses: actions/checkout@v6
82       - uses: actions/setup-python@v6
83         with:
84           python-version: "3.13"
85       - name: Install mypy + typed core deps
86         run: python -m pip install mypy pydantic fastapi uvicorn python-dotenv google-
genai numpy
87       - name: Mypy
88         run: mypy backend training
89
90   # Syntax sweep + import smoke (no GPU stack) ? mirrors ci.yml::import-smoke.
91   # Guards the "SyntaxError / module-level import error shipping while the suite
92   # stays green" class. Deliberately does NOT install cv2/torch/numpy.
93   import-smoke:
94     name: Nightly · Syntax sweep + import smoke (no GPU stack)
95     runs-on: ubuntu-latest
96     timeout-minutes: 10
97     steps:
98       - uses: actions/checkout@v6
99       - uses: actions/setup-python@v6
100         with:
101           python-version: "3.13"

```



```

102         - name: Install minimal deps
103         run: python -m pip install pytest fastapi uvicorn pydantic python-dotenv google-
genai
104         - name: Run import smoke tests
105         run: python -m pytest tests/test_import_smoke.py -v
106
107     # Full offline suite + coverage floor ? mirrors ci.yml::full-suite. CPU-only
108     # torch (no CUDA wheels on a CPU runner), then the editable install pulls the
109     # rest. Floor stays at the ci.yml value (70); never lower it without an owner
110     # decision.
111     full-suite:
112       name: Nightly · Full test suite (offline, mocked)
113       runs-on: ubuntu-latest
114       timeout-minutes: 30
115       steps:
116         - uses: actions/checkout@v6
117         - uses: actions/setup-python@v6
118           with:
119             python-version: "3.13"
120             cache: pip
121         - name: System libs for opencv-python
122           run: |
123             # Drop the runner's pre-installed Microsoft apt repo (azure-cli): it
periodically
124             # 403s on InRelease and aborts `apt-get update` under the `-e` shell, failing
the
125             # build on an unrelated repo we don't use. libgl1/libglb2.0 are in Ubuntu's
archive.
126             sudo rm -f /etc/apt/sources.list.d/*microsoft* /etc/apt/sources.list.d/*azure*
127             sudo apt-get update
128             sudo apt-get install -y libgl1 libglb2.0-0t64 || sudo apt-get install -y
libgl1 libglb2.0-0
129         - name: Install CPU-only torch (no CUDA wheels on a CPU runner)
130         run: python -m pip install torch torchvision --index-url
https://download.pytorch.org/whl/cpu
131         - name: Install trackers (git-only dep ? PyPI wheel is broken)
132         # D13/P12: ByteTrack moved to the standalone `trackers` package (git-only;
133         # see ci.yml for the full rationale). Installed here so the D13/P12 tests
134         # run instead of skipping via pytest.importorskip. Bare version tag (no 'v').
135         run: python -m pip install "trackers @
git+https://github.com/roboflow/trackers.git@2.5.0"
136         - name: Install package (editable) + dev tooling
137         run: python -m pip install -e .[dev]
138         - name: Run full suite (with coverage floor)
139         run: python run_all_tests.py --cov=backend --cov=training --cov-report=term
--cov-fail-under=70
140
141     # Offline quality guard: the committed golden regression fixtures
142     # (docs/GOLDEN_REGRESSION_FIXTURES.md) ? synthetic + de-attributed real
143     # trajectory fixtures, NO video / GPU / DB. Mirrors the offline replay in
144     # ci.yml::golden-crossos (single-OS here; ci.yml already covers cross-OS on
145     # every PR ? nightly is about drift-over-time on the default branch). The
146     # CONTROL replay needs only numpy.
147     golden-fixtures:
148       name: Nightly · Golden fixtures regression (offline quality guard)
149       runs-on: ubuntu-latest
150       timeout-minutes: 15
151       steps:
152         - uses: actions/checkout@v6
153         - uses: actions/setup-python@v6
154           with:
155             python-version: "3.13"
156         - name: Install minimal deps (no GPU stack ? the CONTROL replay needs only numpy)
157         run: python -m pip install pytest fastapi uvicorn pydantic python-dotenv google-
genai numpy

```

```

158     - name: Golden-fixtures characterization (offline)
159     run: |
160         python -m backend.eval.golden_fixtures --check
161         python -m pytest tests/test_golden_fixtures.py
162 tests/test_golden_real_fixtures.py -q
163
164 # Commercial-clean licensing guardrail (MIT/Apache/BSD only) ? mirrors
165 # ci.yml::license-scan. Scans the RUNTIME dependency closure and fails on any
166 # copyleft (GPL/AGPL/LGPL/SSPL). This is exactly the transitive-dep drift the
167 # header cites as a reason the nightly exists ? a new sub-dep can flip a
168 # license between PRs without touching our code.
169 license-scan:
170     name: Nightly · Dependency license scan (no copyleft in runtime deps)
171     runs-on: ubuntu-latest
172     timeout-minutes: 10
173     steps:
174     - uses: actions/checkout@v6
175     - uses: actions/setup-python@v6
176     with:
177         python-version: "3.13"
178     - name: Install runtime deps + pip-licenses
179     run: |
180         python -m pip install .
181         python -m pip install pip-licenses
182     - name: Report dependency licenses
183     run: python -m piplicenses --format=markdown --with-urls --order=license
184     - name: Fail on any copyleft (GPL/AGPL/LGPL/SSPL) in the runtime closure
185     run: |
186         if python -m piplicenses --format=plain | grep -iE "GPL|SSPL"; then
187             echo "::error::A runtime dependency carries a copyleft (GPL/AGPL/LGPL/SSPL)
188             license ? the guardrail is MIT/Apache/BSD only. Review the report above."
189             exit 1
190         fi
191         echo "OK: no GPL/AGPL/LGPL/SSPL in the runtime dependency closure."
192
193 # Built-wheel smoke (install + launch) ? mirrors ci.yml::build-artifact. Builds
194 # the wheel, installs it NON-editable from a non-repo cwd, and launches the
195 # console script. Catches packaging regressions the source suite can't.
196 build-artifact:
197     name: Nightly · Built wheel smoke (install + launch)
198     runs-on: ubuntu-latest
199     timeout-minutes: 20
200     steps:
201     - uses: actions/checkout@v6
202     - uses: actions/setup-python@v6
203     with:
204         python-version: "3.13"
205     - name: System libs for opencv-python
206     run: |
207         # Drop the runner's pre-installed Microsoft apt repo (azure-cli): it
208         periodically
209         # 403s on InRelease and aborts `apt-get update` under the `-e` shell, failing
210         the
211         # build on an unrelated repo we don't use. libgl1/libgl2.0 are in Ubuntu's
212         archive.
213         sudo rm -f /etc/apt/sources.list.d/*microsoft* /etc/apt/sources.list.d/*azure*
214         sudo apt-get update
215         sudo apt-get install -y libgl1 libgl2.0-0t64 || sudo apt-get install -y
216         libgl1 libgl2.0-0
217     - name: Build the wheel
218     run: |
219         python -m pip install build
220         python -m build --wheel
221     - name: Install the built wheel + CPU torch (out-of-band index)
222     run: |

```

```

217 python -m pip install torch torchvision --index-url
https://download.pytorch.org/whl/cpu
218 python -m pip install dist/*.whl
219 - name: Launch the installed console script from a non-repo dir
220 run: |
221 cd /tmp
222 indexer --help
223 python -c "import backend, training, backend.utils.app_home; print('wheel
import OK')"
224
225 # Failure visibility ? a silent nightly is useless. Scheduled runs have no PR
226 # to redden and nobody is watching the Actions tab on a quiet day (the exact
227 # "rot during quiet periods" this workflow exists to catch). On ANY guard
228 # failure, open/refresh a single tracking issue so a human is pinged. Uses the
229 # built-in GITHUB_TOKEN (no secrets); `issues: write` is scoped to THIS job
230 # only ? the guard jobs above keep `contents: read`. Gated on event_name ==
231 # 'schedule': a manual `workflow_dispatch` run is being watched, so no issue.
232 notify-on-failure:
233   name: Nightly · Surface failure (open/refresh tracking issue)
234   needs: [lint, leak-scan, typecheck, import-smoke, full-suite, golden-fixtures,
license-scan, build-artifact]
235   if: failure() && github.event_name == 'schedule'
236   runs-on: ubuntu-latest
237   timeout-minutes: 5
238   permissions:
239     contents: read
240     issues: write
241   steps:
242     - uses: actions/github-script@v7
243       with:
244         script: |
245           const runUrl =
`${context.serverUrl}/${context.repo.owner}/${context.repo.repo}/actions/runs/${context.runId}`;
246           const title = 'Nightly quality regression FAILED';
247           const body = [
248             'The Tier-1 nightly offline quality guard failed.',
249             '',
250             `~ Run: ${runUrl}`,
251             `~ Date (UTC): ${new Date().toISOString}`,
252             '',
253             'One of ruff / leak-scan / mypy / import-smoke / full-suite (cov>=70) /',
254             'golden-fixtures / license-scan / build-artifact regressed.',
255             'Open the run for the failing lane. Close this issue once green.',
256             ].join('\n');
257           // Reuse a single open tracking issue instead of spamming one per night.
258           const existing = await github.rest.issues.listForRepo({
259             owner: context.repo.owner,
260             repo: context.repo.repo,
261             state: 'open',
262             labels: 'nightly-failure',
263             per_page: 1,
264           });
265           if (existing.data.length > 0) {
266             await github.rest.issues.createComment({
267               owner: context.repo.owner,
268               repo: context.repo.repo,
269               issue_number: existing.data[0].number,
270               body,
271             });
272           } else {
273             await github.rest.issues.create({
274               owner: context.repo.owner,
275               repo: context.repo.repo,
276               title,
277               body,

```

```

278         labels: ['nightly-failure'],
279     });
280 }
281

```

8. user (2026-07-02T02:48:00.027Z)

```

{"body": "### Follow-up from #381 review (LOW ? drift)\n\n`nightly.yml` mirrors `ci.yml`'s offline guard lanes by **copying** them, and the coverage floor `70` is a hardcoded literal in both workflows. Nothing enforces that they stay in sync ? `ci.yml` has already recalibrated that floor once, and the next ratchet-up would silently leave the nightly on the stale value.\n\n### Proposed fix\nExtract the shared **offline** guard lanes into a reusable workflow ? e.g. `.github/workflows/_offline-guards.yml` exposing them via `on: workflow_call` (with the coverage floor as an input) ? and have **both** `ci.yml` and `nightly.yml` `uses:` it. That single-sources:\n- the lane set (lint · leak-scan · typecheck · import-smoke · full-suite · golden-fixtures · license-scan · build-artifact), and\n- the coverage floor (one `cov-fail-under` value, passed as an input).\n\n### Why deferred from #381\nThe refactor has to modify `ci.yml` as well, which would break #381's single-file scope. Tracked here per the reviewer's \"fine to ship as-is + file a tracked follow-up\" call.\n\n### Scope / non-goals\n- Offline (Tier-1) lanes only. No GPU / real-video / `run_regression.py`. \n- Keep the two intentional nightly-vs-ci differences: nightly's `golden-fixtures` is single-OS (ci's `golden-crossos` covers the matrix per PR), and nightly's `notify-on-failure` job has no ci equivalent.\n\nRefs: #381, `docs/GOLDEN_REGRESSION_FIXTURES.md`. \n", "state": "OPEN", "title": "ci: single-source the shared offline guard lanes + coverage floor via a workflow_call reusable", "url": "https://github.com/Khelsutra/badminton-highlight-indexer/issues/384"}

```

9. user (2026-07-02T02:48:00.052Z)

```

1      # PEP 621 packaging for the badminton/sports highlight indexer (DEFERRED #7).
2      #
3      # Build backend: hatchling. `backend/` is an intentional NAMESPACE package
4      # (no `backend/__init__.py`; mypy.ini sets explicit_package_bases for the same
5      # reason), so the wheel target packages are listed EXPLICITLY below ? hatchling's
6      # auto-detection would otherwise miss the un-__init__.d `backend` root.
7      #
8      # torch/torchvision are deliberately NOT listed in [project.dependencies]:
9      # their CPU/CUDA wheels need an out-of-band index (--index-url / --extra-index-url),
10     # which PEP 621 cannot express. Install them separately ? see [project.optional-
dependencies]
11     # `gpu` (documentation only) and the README. CI installs CPU torch on its own line.
12     [build-system]
13     requires = ["hatchling"]
14     build-backend = "hatchling.build"
15
16     [project]
17     name = "badminton-highlight-indexer"
18     version = "0.1.0"
19     description = "AI-assisted rally segmentation and highlight indexing for racket sports."
20     readme = "README.md"
21     requires-python = ">=3.10"
22     license = { text = "MIT" }
23     authors = [{ name = "Avi Dullu", email = "avi.dullu@gmail.com" }]
24     keywords = ["badminton", "sports", "video", "highlights", "rally", "segmentation"]
25
26     # Runtime deps mirror requirements.txt, EXCEPT torch/torchvision (see header).
27     dependencies = [
28         "fastapi>=0.111.0",
29         "uvicorn>=0.30.1",
30         "opencv-python>=4.9.0.80",
31         "numpy>=1.26.4",
32         "pydantic>=2.7.1",
33         "jinja2>=3.1.4",
34         "python-dotenv>=1.0.0",
35         "google-genai>=2.4.0",
36         "supervision>=0.21.0",

```

```

37         # Used directly on the serving path (backend/features/rally_seq.py: uniform_filter1d
/
38         # maximum_filter1d); previously present only TRANSITIVELY via supervision, so a
supervision
39         # bump or a pruned-transitive freeze could silently break inference. Declare it.
(BSD-3.)
40         "scipy>=1.10",
41         # D13/P12: ByteTrack moved to the standalone `trackers` package (Apache-2.0). Its
42         # PyPI wheel is broken (metadata-only), so it MUST be installed from git ? see
43         # requirements.txt. PEP 621 cannot express a git URL, so it's NOT listed here
44         # (same pattern as torch/torchvision). CI installs it on its own line.
45     ]
46
47     [project.optional-dependencies]
48     # Test + lint/type tooling ? matches the CI lanes (.github/workflows/ci.yml).
49     dev = [
50         "pytest",
51         "pytest-cov",
52         "httpx",          # fastapi TestClient transport
53         "ruff",
54         "mypy",
55         "PyJWT[crypto]>=2.8", # M9 edge-auth: Cf-Access JWT verify (RS256) ? tests + CI
56     ]
57
58     # M9 edge-auth (Cloudflare Access). DEFAULT-OFF: only needed when
security.edge_auth_enabled=True.
59     # `jwt` is lazy-imported, so the base runtime works without this; install it for
hosted/multi-tenant.
60     auth = ["PyJWT[crypto]>=2.8"]
61
62     # Per-video observability reports (SOFT dependency: the pipeline degrades to a
63     # no-op if absent). Pinned to its git tag per BACKLOG. The private repo is
64     # fetched over SSH; if you lack access, simply omit this group ? reporting
65     # tests skip via pytest.importorskip. For local dev against a sibling checkout:
66     # pip install -e ../sports-obsreport --no-build-isolation
67     reporting = [
68         "obsreport @ git+ssh://git@github.com/avidullu/sports-obsreport.git@v0.1.3",
69     ]
70
71     # `gpu` is DOCUMENTATION ONLY ? it is intentionally empty. torch/torchvision
72     # CUDA wheels require `--index-url`/`--extra-index-url`, which PEP 621 cannot
73     # express, so they are NOT declarable here. Install them out-of-band instead:
74     # # CUDA 12.4 box:
75     # pip install torch>=2.12.0 torchvision>=0.27.0 --index-url
https://download.pytorch.org/whl/cu124
76     # # CPU-only box (also what CI uses):
77     # pip install torch torchvision --index-url https://download.pytorch.org/whl/cpu
78     gpu = []
79
80     # Storage backends (docs/DECOUPLED_COMPUTE/). The default `local` backend needs NONE of
81     # these. Cloud SDKs are imported lazily, so `import backend.storage` works without them;
82     # install the extra for the bucket you use. (boto3's S3 client covers AWS + Cloudflare
R2 +
83     # DO Spaces + MinIO via endpoint_url.) `gdrive://` needs NO extra ? it reads a locally
84     # MOUNTED Google Drive (Google Drive for Desktop) as plain files; see
backend/storage/gdrive.py.
85     storage-s3 = ["boto3>=1.34"]
86     storage-gcs = ["google-cloud-storage>=2.16"]
87
88     [project.scripts]
89     # `indexer --server`, `indexer --help`, etc. -> backend/main.py::main()
90     indexer = "backend.main:main"
91
92     [tool.hatch.metadata]
93     # The `reporting` extra pins obsreport to a git tag (a PEP 508 *direct

```

```

94 # reference*); hatchling requires opting in to allow those in metadata.
95 allow-direct-references = true
96
97 [tool.hatch.build.targets.wheel]
98 # `backend` is a namespace package (no __init__.py); list both import roots
99 # explicitly so an editable/wheel install exposes `import backend` and
100 # `import training` regardless of cwd.
101 packages = ["backend", "training"]
102

```

10. user (2026-07-02T02:48:00.421Z)

```

1  #!/usr/bin/env python
2  """Run the FULL test suite for the badminton-highlight-indexer.
3
4  A single, discoverable entry point so you (or a reviewer / CI) can verify the
5  whole suite before opening a PR:
6
7      python run_all_tests.py          # full suite, quiet
8      python run_all_tests.py -v       # verbose (passthrough to pytest)
9      python run_all_tests.py -k config # filter by keyword (passthrough)
10     python run_all_tests.py --cov=backend --cov-report=term-missing # coverage
11
12  Anything after the script name is passed straight through to pytest.
13
14  Notes
15  -----
16  - The suite is fully mocked (no GPU, no network, no WSL) and runs offline.
17  - The observability **reporting** tests skip cleanly if the optional ``obsreport``
18    package isn't installed (they use ``pytest.importorskip``). To exercise them too:
19      pip install -e ../sports-obsreport --no-build-isolation
20  - Exit code is pytest's own, so this is CI / pre-PR friendly (0 = all green).
21  """
22
23  from __future__ import annotations
24
25  import importlib.util
26  import subprocess
27  import sys
28
29
30  def _installed(module: str) -> bool:
31      try:
32          return importlib.util.find_spec(module) is not None
33      except (ImportError, ValueError):
34          return False
35
36
37  def main() -> int:
38      args = list(sys.argv[1:])
39      # Default to quiet unless the caller chose a verbosity.
40      if not any(a in ("-q", "-v", "-vv", "-vvv", "--quiet", "--verbose") for a in args):
41          args = ["-q", *args]
42      cmd = [sys.executable, "-m", "pytest", *args]
43
44      obs = (
45          "available"
46          if _installed("obsreport")
47          else "NOT installed ? reporting tests will SKIP"
48      )
49      bar = "=" * 72
50      print(bar)
51      print("Full test suite: " + " ".join(cmd))
52      print(f"obsreport (optional reporting dep): {obs}")
53      print(bar, flush=True)

```

```

54
55         rc = subprocess.run(cmd).returncode
56
57         print(bar)
58         print(
59             " RESULT: "
60             + ("PASS - all tests green" if rc == 0 else f"FAIL - pytest exit {rc}")
61         )
62         print(bar)
63         return rc
64
65
66     if __name__ == "__main__":
67         raise SystemExit(main())
68

```

11. user (2026-07-02T02:48:18.375Z)

```

pytest.ini
.coveragerc
mypy.ini
ruff.toml

```

12. user (2026-07-02T02:48:19.788Z)

```

tests\test_api_cloud_dispatch.py:1
tests\test_async_jobs.py:1
tests\test_app_home.py:1
tests\test_api_annotations.py:1
tests\test_api_assets.py:1
tests\test_api_endpoints.py:2
tests\test_ablation.py:6
tests\test_bind_and_host.py:1
tests\test_config_deployment.py:1
tests\test_edge_auth.py:2
tests\test_import_smoke.py:2
tests\test_job_worker_loop.py:1
tests\test_nightly_reelcount.py:1
tests\test_person_detector.py:3
tests\test_rally_seg_eval.py:1
tests\test_reporting.py:1
tests\test_reporting_harvest.py:1
tests\test_reporting_spec_faq.py:1
tests\test_served_gate_a_regression.py:2
tests\test_ui_bundle.py:1
tests\test_wasb_cache.py:5
tests\test_upload_download.py:1

```

Found 37 total occurrences across 22 files.

13. user (2026-07-02T02:48:25.094Z)

```

1     [pytest]
2     # When `pytest` is invoked without an explicit path, only collect from tests/.
3     # scratch/ holds debugging scripts that hit real APIs (Gemini upload, etc.)
4     # and should never run as part of the regular test suite.
5     testpaths = tests
6

```

14. user (2026-07-02T02:48:25.550Z)

```

1     [run]
2     # Excluded from coverage: the legacy motion segmenter (demo baseline with
3     # fabricated metadata) and the concrete validators (thin OpenCV/ffprobe

```

```

4      # wrappers only exercisable against real media files). validators/base.py
5      # stays counted ? it has a dedicated unit suite.
6      omit =
7          backend/pipeline/segmenters/motion.py
8          backend/pipeline/validators/format.py
9          backend/pipeline/validators/resolution_fps.py
10         backend/pipeline/validators/pts_continuity.py
11         backend/pipeline/validators/scene_cut.py
12         backend/pipeline/validators/blur.py
13         backend/pipeline/validators/relevance.py
14

```

15. user (2026-07-02T02:48:26.505Z)

```

1      # Lint-only lane (no `ruff format` enforcement ? a whole-repo reformat would
2      # bury blame history for zero behavior value).
3      #
4      # Selected: pyflakes + pycodestyle errors (ruff defaults) + pycodestyle
5      # warnings (W) + bugbear (B) + isort (I ? consistent import ordering).
6      target-version = "py38"
7
8      # scratch/ = throwaway debugging scripts that may hit real APIs; it is also
9      # excluded from pytest collection (see pytest.ini).
10     extend-exclude = ["scratch"]
11
12     [lint]
13     extend-select = ["W", "B", "I"]
14
15     [lint.per-file-ignores]
16     # Entrypoint scripts deliberately run setup (load_dotenv, logging config,
17     # sys.path) before importing backend modules ? import order is load-bearing.
18     "backend/main.py" = ["E402"]
19     "scripts/run_phase3_eval.py" = ["E402"]
20     "scripts/run_process_and_stitch.py" = ["E402"]
21

```

16. user (2026-07-02T02:48:26.814Z)

```

1      [mypy]
2      # Lenient, GREEN baseline (quality sweep AI-4): the gate protects the clean
3      # modules and pins an explicit quarantine list below ? not a typing crusade.
4      # Ratchet: when you touch a quarantined module, try removing its entry.
5      ignore_missing_imports = True
6      # backend/ is a namespace package (no root __init__.py) ? required so mypy
7      # maps files to backend.* module names instead of duplicating top-level names.
8      explicit_package_bases = True
9
10     # --- Quarantine (known-loose modules, each with a reason) ---
11
12     # Optional module globals (db_instance/global_config set by create_app() / main()).
13     # Route handlers use request.app.state exclusively; helpers take explicit params.
14     # backend.api.job_worker lifted from quarantine (now takes explicit config/db params).
15     [mypy-backend.main]
16     ignore_errors = True
17
18     # Worker dispatcher: composes the SAME validator/segmenter/report building blocks as
19     # backend.main, which are typed `config: dict` but receive the AppConfig migration shim
20     # at runtime (the .get() compat layer). Shares backend.main's config-shim quarantine
21     # until the typed-config migration removes the dict?AppConfig duality everywhere.
22     [mypy-backend.worker.__main__]
23     ignore_errors = True
24
25     # yolo_hybrid: the fusion-P1 extraction moved its loose torchvision/cv2 inline code
26     # to detectors/person_detector.py (which IS type-clean, no quarantine). What remains
27     # here is pre-existing orchestration debt unrelated to that move (implicit-Optional

```



```

28     # process_video defaults, the Success|Failure ABC-override mismatch, a Future[...]
29     # annotation) ? ratchet those out in a focused typing pass, not here.
30     # (wasb_hybrid/tracknet_hybrid left the quarantine with the AI-7 collapse.)
31     [mypy-backend.pipeline.segmenters.yolo_hybrid]
32     ignore_errors = True
33
34     [mypy-backend.pipeline.segmenters.gemini]
35     ignore_errors = True
36
37     # Registry builds cls(None, {}) probes; typing it honestly needs a Protocol.
38     [mypy-backend.pipeline.segmenters.base]
39     ignore_errors = True
40

```

17. user (2026-07-02T02:48:47.345Z)

```

tests\test_ablation.py:400:     pytest.importorskip("reportlab")
tests\test_ablation.py:401:     pypdf = pytest.importorskip("pypdf")
tests\test_ablation.py:525:@pytest.mark.skipif(
tests\test_ablation.py:555:@pytest.mark.skipif(
tests\test_ablation.py:629:     pytest.importorskip("reportlab")
tests\test_ablation.py:630:     pytest.importorskip("pypdf")
tests\test_api_endpoints.py:11:pytest.importorskip("httpx") # TestClient needs httpx; skip
cleanly where absent
tests\test_api_endpoints.py:889:     pytest.importorskip("obsreport") # the report writer is a
no-op without it
tests\test_api_assets.py:14:pytest.importorskip("httpx") # TestClient needs httpx; skip cleanly
where absent
tests\test_api_annotations.py:7:pytest.importorskip("httpx") # TestClient needs httpx; skip
cleanly where absent
tests\test_async_jobs.py:20:pytest.importorskip("httpx") # TestClient needs httpx; skip cleanly
where absent
tests\test_app_home.py:193:         pytest.skip("Windows drive-letter semantics only meaningful
on nt")
tests\test_api_cloud_dispatch.py:16:pytest.importorskip("httpx")
tests\test_bind_and_host.py:5:pytest.importorskip("httpx") # TestClient needs httpx
tests\test_config_deployment.py:401:         pytest.skip("no repo config.json")
tests\test_edge_auth.py:14:pytest.importorskip(
tests\test_edge_auth.py:218:pytest.importorskip("httpx") # fastapi TestClient transport
tests\test_import_smoke.py:110:     pytest.importorskip("numpy")
tests\test_import_smoke.py:111:     pytest.importorskip("scipy")
tests\test_job_worker_loop.py:26:pytest.importorskip("httpx")
tests\test_nightly_reelcount.py:276:     pytest.importorskip("cv2")
tests\test_person_detector.py:215:     pytest.importorskip("trackers")
tests\test_person_detector.py:270:     pytest.importorskip("trackers")
tests\test_person_detector.py:290:     pytest.importorskip("trackers")
tests\test_rally_seg_eval.py:124:     pytest.importorskip(
tests\test_reporting.py:80:# (see test_reporting_harvest.py which uses pytest.importorskip).
Those tests
tests\test_reporting_spec_faq.py:14:pytest.importorskip("obsreport")
tests\test_reporting_harvest.py:11:pytest.importorskip("obsreport")
tests\test_served_gate_a_regression.py:47:pytest.importorskip("cv2")
tests\test_served_gate_a_regression.py:82:_skip = pytest.mark.skipif(
tests\test_ui_bundle.py:8:pytest.importorskip("httpx")
tests\test_upload_download.py:16:pytest.importorskip("httpx") # TestClient needs httpx; skip
cleanly where absent
tests\test_wasb_cache.py:368:     cv2 = pytest.importorskip("cv2") # noqa: F841 (importorskip
gates the suite on cv2)
tests\test_wasb_cache.py:375:     cv2 = pytest.importorskip("cv2")
tests\test_wasb_cache.py:401:     pytest.importorskip("cv2")
tests\test_wasb_cache.py:418:     cv2 = pytest.importorskip("cv2")
tests\test_wasb_cache.py:419:     Image = pytest.importorskip("PIL.Image")

```

18. user (2026-07-02T02:48:59.066Z)

```

385         granularity="group",
386     ),
387 )
388 out = ablation.format_table(report)
389 assert "Ablation report" in out
390 assert "?F1(full?ablated)" in out
391 assert "verdict" in out
392 # the highest-?F1 signal name appears before the lowest in the rendered order
393 top = report.sorted_rows()[0].signal
394 bottom = report.sorted_rows()[-1].signal
395 assert out.index(top) < out.index(bottom)
396
397
398 def test_export_pdf_is_valid_nontrivial(tmp_path):
399     """A valid PDF with at least one page is produced, and it contains the verdict
legend."""
400     pytest.importorskip("reportlab")
401     pypdf = pytest.importorskip("pypdf")
402     videos = _learned_videos(6)
403     report = run_ablation(
404         videos,
405         AblationConfig(
406             full=ablation.learned_full_variant(),
407             signals=ablation.default_signals(),
408             granularity="group",
409         ),
410     )
411     out = str(tmp_path / "ablation.pdf")
412     ablation.export_pdf(report, out)
413     assert os.path.isfile(out)
414     assert os.path.getsize(out) > 1000 # non-trivial
415     reader = pypdf.PdfReader(out)
416     assert len(reader.pages) > 0
417     text = "".join(p.extract_text() for p in reader.pages)
418     assert "Ablation" in text
419     assert "clears-bar" in text # the verdict key rendered
420
421
422 # ----- corpus loader
(no GPU)
423
424
425 def test_load_videos_tolerates_missing_audio(tmp_path):
426     """An older feature JSON without an ``audio`` block still loads ? audio columns
default to
427     0.0 (loader never crashes on a pre-audio JSON)."""
428     cand = {
429         "start": 0.0,
430         "end": 10.0,
431         "shuttle": {fn: 1.0 for fn in ablation.SHUTTLE_COLUMNS},
432         "person": {fn: 0.0 for fn in ablation.PERSON_COLUMNS},
433         "label": 1,
434     }
435     data = {
436         "name": "noaudio",
437         "fps": 30.0,
438         "frame_width": 1920.0,
439         "candidates": [cand],
440         "gts": [[0.0, 10.0]],
441     }
442     with open(tmp_path / "noaudio_golden_features.json", "w", encoding="utf-8") as f:
443         json.dump(data, f)
444     videos = ablation.load_videos(str(tmp_path))

```

```

515         cands = json.load(f).get("candidates") or []
516     except (OSError, ValueError):
517         continue
518     if _is_fusion_schema(
519         cands
520     ): # canonical predicate (SSOT in backend.eval.experiment, #215)
521         n += 1
522     return n >= 6
523
524
525     @pytest.mark.skipif(
526         not _golden_present(),
527         reason="golden output/*_golden_features.json absent (run fusion_golden first)",
528     )
529     def test_litmus_reproduces_gate_a():
530         """LITMUS: on the 6 real golden feature JSONs the runner reproduces the hand-
531         measured Gate-A
532         result ? marginal ? F1 ? +0.074, 6/6 wins, p ? 0.031, CI ? [+0.042, +0.104], within
533         tolerance.
534         This validates the whole framework against the recorded ground truth. Skips cleanly
535         in CI
536         where the golden substrate is absent; runs on the owner box where it is present."""
537         videos = ablation.load_videos(GOLDEN_DIR)
538         assert len(videos) == 6
539         cfg = ablation.litmus_config()
540         report = run_ablation(videos, cfg)
541         assert len(report.rows) == 1
542         row = report.rows[0]
543         assert row.signal == "gate_a"
544         # The recorded QUALITY_ITERATIONS truth (2026-06-14), reproduced within tolerance.
545         assert row.delta_f1 == pytest.approx(0.074, abs=0.005)
546         assert row.sign_wins == 6 and row.sign_losses == 0
547         assert row.sign_p_value == pytest.approx(0.031, abs=0.005)
548         assert row.ci_low == pytest.approx(0.042, abs=0.01)
549         assert row.ci_high == pytest.approx(0.104, abs=0.01)
550         assert row.ci_excludes_zero is True
551         # Gate-A is structural ? reported but never auto-promoted to earns_keep.
552         assert row.verdict == ablation.STRUCTURAL_PROTECTED
553         # C4: over-segmentation drops 1.68 ? 1.01 when Gate-A is on (full).
554         assert row.seg_ratio_full == pytest.approx(1.01, abs=0.05)
555         assert row.seg_ratio_ablated == pytest.approx(1.68, abs=0.05)
556
557     @pytest.mark.skipif(
558         not _golden_present(), reason="golden output/*_golden_features.json absent"
559     )
560     def test_litmus_report_is_well_formed():
561         """The litmus report is a proper AblationReport (provenance + a sorted single
562         row)."""
563         report = run_ablation(ablation.load_videos(GOLDEN_DIR), ablation.litmus_config())
564         assert isinstance(report, AblationReport)
565         assert report.num_videos == 6
566         assert report.full_set_signals == ["gate_a"]
567         assert report.label_set_hash
568         out = ablation.format_table(report)
569         assert "gate_a" in out
570
571     # ----- edge
572     cases
573
574     def test_min_players_gate_knob_zeroed():
575         """The ``min_players`` gate knob is zeroed on ablation (the Gate-B analog)."""
576         full = Variant(name="full", min_players=2, feature_names=["a"])

```

20. user (2026-07-02T02:49:01.408Z)

```
40     import json
41     import os
42
43     import pytest
44
45     # cv2 is imported transitively by trajectory_hybrid (the served engine). On a box
without it
46     # (CI), skip the whole module rather than erroring at import.
47     pytest.importorskip("cv2")
48
49     from backend.eval import segmentation_metrics as _segm # noqa: E402
50     from backend.eval import served_gate_a as sga # noqa: E402
51     from backend.eval.calibrate_wasb import load_golden_gts # noqa: E402
52     from backend.eval.metrics import score # noqa: E402
53     from backend.pipeline.detectors import rally_gate # noqa: E402
54     from backend.pipeline.detectors.tracknet_runner import TrackNetRunner # noqa: E402
55
56     # The golden manifest lives in the MAIN checkout's output/ (gitignored; absent in a
fresh
57     # worktree / CI). Resolve it relative to this repo root first, with an env override for
an
58     # out-of-tree checkout.
59     _REPO_ROOT = os.path.dirname(os.path.dirname(os.path.abspath(__file__)))
60     MANIFEST = os.environ.get(
61         "SERVED_GATE_A_MANIFEST",
62         os.path.join(_REPO_ROOT, "output", "human_lovo_manifest.json"),
63     )
64
65
66     def _golden_specs():
67         """Return the manifest specs whose trajectory + label files are all present, else
[]."""
68         if not os.path.isfile(MANIFEST):
69             return []
70         with open(MANIFEST, encoding="utf-8") as f:
71             specs = json.load(f)
72         ok = [
73             s
74             for s in specs
75             if os.path.isfile(str(s.get("trajectory", "")))
76             and os.path.isfile(str(s.get("labels", "")))
77         ]
78         return ok if len(ok) >= 6 else []
79
80
81     _SPECS = _golden_specs()
82     _skip = pytest.mark.skipif(
83         not _SPECS,
84         reason=f"golden trajectories absent (manifest {MANIFEST!r}); runs on the owner box",
85     )
86
87
88     # ---- #398 regression: golden-free (runs in CI; only the golden DATA is absent there,
not cv2) ----
89     def test_served_idx_cfg_is_typed_and_seams_accept_it(tmp_path):
90         """The served litmus must pass a TYPED IndexingConfig (what production passes), not
a dict ?
91         `_enrich_candidate_stats` reads `idx_cfg.fusion`, so a dict raised AttributeError
(#398). Build
92         the config + drive the full served seam path on a synthetic trajectory (no golden
data needed)."""
93         from backend.config.models import IndexingConfig
94
```

```

95         cfg = sga._served_idx_cfg(2)
96         assert isinstance(cfg, IndexingConfig)
97         assert cfg.fusion.min_crossings == 2
98         traj = tmp_path / "synth.csv"
99         rows = ["Frame,Visibility,X,Y"]

```

21. user (2026-07-02T02:49:02.731Z)

```

docs\ABLATION_LAYER.md:61:reportlab report (table of marginal contribution + CI + sign test + a
verdict column).
backend\eval\ablation.py:527:# The ~116-line reportlab PDF builder lives in a sibling module to
keep this file a pure driver;
backend\eval\ablation_pdf.py:1:~~~~~reportlab PDF renderer for the ablation report (split out of
`backend.eval.ablation`~).
backend\eval\ablation_pdf.py:4:booktabs-style PDF. `reportlab` is imported lazily inside
`export_pdf` so importing this
backend\eval\ablation_pdf.py:8:Kept as its own module so the ~116-line reportlab builder doesn't
bloat the pure-driver
backend\eval\ablation_pdf.py:28: ~~~~~Render the `AblationReport` to a clean reportlab PDF: a
table of each signal's marginal
backend\eval\ablation_pdf.py:32: Returns the written path. `reportlab` is an offline, pure-
Python dependency (no GPU).~~~~~
backend\eval\ablation_pdf.py:33: from reportlab.lib import colors
backend\eval\ablation_pdf.py:34: from reportlab.lib.enums import TA_CENTER, TA_LEFT
backend\eval\ablation_pdf.py:35: from reportlab.lib.pagesizes import A4
backend\eval\ablation_pdf.py:36: from reportlab.lib.styles import ParagraphStyle,
getSampleStyleSheet
backend\eval\ablation_pdf.py:37: from reportlab.lib.units import cm
backend\eval\ablation_pdf.py:38: from reportlab.platypus import (

```

22. user (2026-07-02T02:49:31.699Z)

```

completed      failure refactor(D1/P10): extract orchestration.py from main.py (god-module
split, step 1)    CI      refactor/p10-split-main pull_request      28244991823      3m28s
2026-06-26T14:37:23Z
completed      failure refactor(D1/P10): extract orchestration.py from main.py (god-module
split, step 1)    CI      refactor/p10-split-main pull_request      28244470989      2m52s
2026-06-26T14:28:10Z
completed      failure chore: ruff format the whole repo (259 files, formatting-only) CI
chore/ruff-format-repo pull_request      28238584230      2m49s 2026-06-26T12:39:44Z
completed      failure feat(D13/P12): migrate ByteTrack to standalone trackers package CI
d13-p12/bytetrack-migration pull_request      28237503425      2m57s 2026-06-26T12:18:03Z
completed      failure feat(D13/P12): migrate ByteTrack to standalone trackers package CI
d13-p12/bytetrack-migration pull_request      28232820180      1m16s 2026-06-26T10:37:29Z
completed      failure feat(D13/P12): migrate ByteTrack to standalone trackers package CI
d13-p12/bytetrack-migration pull_request      28232698842      1m20s 2026-06-26T10:34:46Z
completed      failure feat(D13/P12): migrate ByteTrack to standalone trackers package CI
d13-p12/bytetrack-migration pull_request      28191091090      3m5s 2026-06-25T18:16:28Z
completed      failure chore: ruff format gemini_refine.py + trajectory_hybrid.py CI
chore/format-gemini-refine-trajectory-hybrid pull_request      28183805969      2m35s
2026-06-25T16:08:07Z
completed      failure fix(tests): resolve remaining test signature mismatches from app-factory
refactor CI      fix/remaining-test-sigs pull_request      28093413101      2m47s
2026-06-24T10:53:35Z
completed      failure refactor(main): app-factory + typed-config migration (D2/D3/P2/P3/P5/?
CI master push      28092411029      2m34s 2026-06-24T10:33:47Z
completed      failure refactor(main): app-factory + typed-config migration (D2/D3/P2/P3/P5/P7)
CI fix/app-factory-refactor-d2 pull_request      28091928133      2m52s
2026-06-24T10:24:39Z
completed      failure refactor(main): app-factory + typed-config migration (D2/D3/P2/P3/P5/P7)
CI fix/app-factory-refactor-d2 pull_request      28091694811      2m47s
2026-06-24T10:20:11Z
completed      failure refactor: segmenters use typed AppConfig access (D3-b) CI
refactor/segmenters-typed-config pull_request      28060809741      2m28s
2026-06-23T22:16:04Z

```

```
completed      failure refactor: segmenters use typed AppConfig access (D3-b) CI
refactor/segmenters-typed-config      pull_request      28059280286      2m23s
2026-06-23T21:46:22Z
completed      failure fix: segmenter_must_be_registered validator actually validates (B5)
CI      fix/segmenter-validator-actually-validates      pull_request      28052756302      2m43s
2026-06-23T19:51:33Z
---
```

23. user (2026-07-02T02:49:33.094Z)

```
config.json
---
af1ccbe chore(#399): launch-PR process hardening ? link-check CI, PR template, doc cross-links
(#402)
ffb1f53 ci: harden the 'System libs' apt step against the runner's Microsoft-repo 403 flake
(#391)
---
/c/Users/avidu/AppData/Local/Programs/Python/Python313/Scripts/ruff
/c/Users/avidu/AppData/Local/Programs/Python/Python313/Scripts/mypy
ruff 0.15.16
mypy 2.1.0 (compiled: yes)
```

24. user (2026-07-02T02:49:33.112Z)

```
390         assert saved["deployment"] == {"profile": "", "compute_target": ""}
391         cfg = load_config(str(p))
392         assert cfg.deployment.profile == ""
393
394
395     def test_repo_config_json_has_no_profile():
396         """The shipped config.json must not silently opt into a profile (default path stays
today's
397         behaviour for the real deployment)."""
398         repo_root = os.path.dirname(os.path.dirname(os.path.abspath(__file__)))
399         cfg_path = os.path.join(repo_root, "config.json")
400         if not os.path.exists(cfg_path):
401             pytest.skip("no repo config.json")
402         cfg = load_config(cfg_path)
403         assert cfg.deployment.profile == ""
404
```

25. user (2026-07-02T02:49:33.273Z)

```
tests/test_reporting_harvest.py:14
tests/test_reporting_spec_faq.py:2
tests/test_reporting.py:5
```

26. user (2026-07-02T02:49:53.578Z)

All checks passed!

27. user (2026-07-02T02:49:53.975Z)

```
tests\test_served_gate_a_regression.py:5

Found 5 total occurrences across 1 file.
```

28. user (2026-07-02T02:50:12.974Z)

```
backend\pipeline\validators\base.py:44: note: By default the bodies of untyped functions are not
checked, consider using --check-untyped-defs [annotation-unchecked]
backend\sports\_init_.py:9: note: By default the bodies of untyped functions are not checked,
consider using --check-untyped-defs [annotation-unchecked]
backend\storage\gcs.py:31: error: Module "google.cloud" has no attribute "storage" [attr-
```

```
defined]
backend\providers\__init__.py:9: note: By default the bodies of untyped functions are not
checked, consider using --check-untyped-defs [annotation-unchecked]
Found 1 error in 1 file (checked 141 source files)
```

29. user (2026-07-02T02:50:34.697Z)

```
20         self,
21         *,
22         client: Any = None,
23         project: Optional[str] = None,
24         signed_url_ttl: int = 3600,
25         **_ignored,
26     ) -> None:
27         self._ttl = signed_url_ttl
28         if client is not None:
29             self._client = client
30         return
31         from google.cloud import (
32             storage, # type: ignore # lazy: only when a gcs:// URI is used
33         )
34
35         self._client = storage.Client(project=project)
36
37     def _blob(self, ref: StorageRef):
38         return self._client.bucket(ref.bucket).blob(ref.key)
39
```

30. user (2026-07-02T02:50:49.151Z)

```
tests/test_yolo_hybrid.py::test_yolo_hybrid_store_candidates_disabled
tests/test_yolo_hybrid.py::test_yolo_chunk_dirs_follow_output_dir
```

1558 tests collected in 12.15s

31. user (2026-07-02T02:51:08.648Z)

```
118         agg = aggregate([])
119         assert agg["n"] == 0 and agg["f1"] is None
120
121
122     # --- W2: net-crossings gate (Gate-A) on the windowing -----
123     #
124     def test_windows_for_netcross_gate(tmp_path):
125         pytest.importorskip(
126             "cv2"
127         ) # rally_gate import path tolerance (no cv2 actually needed)
128         video = {
129             "trajectory": _write_oscillating_traj(tmp_path / "osc.csv"),
130             "fps": 30.0,
131             "frame_width": 1280.0,
132         }
```

32. user (2026-07-02T02:51:11.155Z)

```
{
  "checks": [
    {
      "conclusion": "SUCCESS",
      "name": "Ruff lint",
    },
    {
      "conclusion": "SUCCESS",
      "name": "Leak scan (PII / attribution guard)",
    },
    {
      "conclusion": "SUCCESS",
      "name": "Mypy (lenient baseline)",
    },
    {
      "conclusion": "SUCCESS",
      "name": "Syntax sweep + import smoke (no GPU stack)",
    },
    {
      "conclusion": "FAILURE",
      "name": "Full test suite (offline, mocked)",
    },
    {
      "conclusion": "SUCCESS",
      "name": "Golden fixtures cross-OS (ubuntu-latest)",
    },
    {
      "conclusion": "SUCCESS",
      "name": "Golden fixtures cross-OS (windows-latest)",
    },
    {
      "conclusion": "SUCCESS",
      "name": "Golden fixtures cross-OS (macos-latest)",
    },
    {
      "conclusion": "SUCCESS",
      "name": "Dependency license scan (no copyleft in runtime deps)",
    },
    {
      "conclusion": "SUCCESS",
      "name": "Built wheel smoke (install +
```

```
launch)"}], "state": "OPEN", "title": "refactor(D1/P10): extract orchestration.py from main.py (god-  
module split, step 1)"}]
```

33. assistant (2026-07-02T02:52:23.429Z)

All checks complete. I have a full picture of the engine's CI + test health. Compiling the findings.

34. user (2026-07-02T02:53:14.185Z)

Structured output provided successfully